PERVERSE COHOMOLOGY OF SPHERICAL PARITY COMPLEXES IS TILTING IN ARBITRARY COEFFICIENT CHARACTERISTIC

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Editorial note. Lemma4.13 explicitly invokes the classical rank-one parity-preserving convolution calculation and parity-complex classification in JMW2014/Springer1982. This precise foundational result is allowed as stated, not claimed to be fully reproved in the short source paragraph. The new equivariant-summand construction and flag/Grassmannian transport remain author obligations.

Editorial note. Lemma4.8 obtains a cohomological shift of E_lambda as a summand; shifting the constructed equivariant parity object back gives the unshifted formulation because induction, forgetful functors and parity are shift-compatible. No good-characteristic assumption is added.

Allowed standard prerequisites (editorial).

- Six-functor formalism, smooth base change, Verdier duality, perverse t-structures and highest weight categories. General sheaf/category infrastructure, including field-coefficient perverse convolution. The Iwahori–Whittaker equivalence and its modular-coefficient proof are not supplied.
- Geometric Casselman–Shalika formula over characteristic-zero coefficients and dimensions of semi-infinite intersections. Existing foundational input; descent of top-cohomology vanishing to modular coefficients, adjunction and the equivalence proof remain obligations.
- Foundational parity-sheaf existence/uniqueness and rank-one convolution methods. Classification and standard rank-one parity-preservation arguments only. The current equivariant direct-summand lift, IW parity-versus-tilting identification and passage through affine flags/Grassmannians remain the authored local proof. Lemma4.13 explicitly refers its rank-one calculation to these standard arguments; independent completeness review must accept that precise scope.

2. Constructible sheaves on affine Grassmannians and affine flag varieties

2.1. **Notation.** Let $\mathbb{F}$ be an algebraically closed field of characteristic $p > 0$. Let G be a connected reductive algebraic group over $\mathbb{F}$, let $B^- \subset G$ be a Borel subgroup, and let $T \subset B^-$ be a maximal torus. Let also $B^+ \subset G$ be the Borel subgroup opposite to B^- (with respect to T), and let U^+ be its unipotent radical.

We denote by $\boldsymbol{X} := X^*(T)$ the character lattice of T, by $\boldsymbol{X}^\vee := X_*(T)$ its cocharacter lattice, by $\Delta \subset \boldsymbol{X}$ the root system of (G, T), and by $\Delta^\vee \subset \boldsymbol{X}^\vee$ the corresponding coroots. We choose the system of positive roots $\Delta^+ \subset \Delta$ consisting of the T-weights in $\mathrm{Lie}(U^+)$, and denote by $\boldsymbol{X}^\vee_+ \subset \boldsymbol{X}^\vee$, resp. $\boldsymbol{X}^\vee_{++} \subset \boldsymbol{X}^\vee$ the corresponding subset of dominant cocharacters, resp. of strictly dominant cocharacters.

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We also denote by $\Delta_{\mathrm{s}} \subset \Delta$ the corresponding subset of simple roots, and set

$$\rho = \frac{1}{2} \sum_{\alpha \in \Delta^+} \alpha \quad \in \mathbb{Q} \otimes_{\mathbb{Z}} \boldsymbol{X}.$$

For any $\alpha \in \Delta_{\mathrm{s}}$ we choose an isomorphism between the additive group $\mathbb{G}_{\mathrm{a}}$ and the root subgroup U_α of G associated with α, and denote it u_α.

We will assume[1] that there exists $\varsigma \in \boldsymbol{X}^\vee$ such that $\langle \varsigma, \alpha \rangle = 1$ for any $\alpha \in \Delta_{\mathrm{s}}$; then we have $\boldsymbol{X}^\vee_{++} = \boldsymbol{X}^\vee_+ + \varsigma$. (Such a cocharacter might not be unique; we fix a choice once and for all.)

Let W_{f} be the Weyl group of (G, T), and let $W := W_{\mathrm{f}} \ltimes \boldsymbol{X}^\vee$ be the corresponding (extended) affine Weyl group. For $\lambda \in \boldsymbol{X}^\vee$ we will denote by t_λ the associated element of W. If $w \in W$ and $w = t_\lambda v$ with $\lambda \in \boldsymbol{X}^\vee$ and $v \in W_{\mathrm{f}}$, we set

$$\ell(w) = \sum_{\substack{\alpha \in \Delta^+ \\ v(\alpha) \in \Delta^+}} |\langle \lambda, \alpha \rangle| + \sum_{\substack{\alpha \in \Delta^+ \\ v(\alpha) \in -\Delta^+}} |1 + \langle \lambda, \alpha \rangle|.$$

The restriction of ℓ to the semi-direct product W^{Cox} of W_{f} with the coroot lattice is the length function for a natural Coxeter group structure, and if we set $\Omega := \{w \in W \mid \ell(w) = 0\}$ then multiplication induces a group isomorphism $W^{\mathrm{Cox}} \rtimes \Omega \xrightarrow{\sim} W$.

2.2. **The affine Grassmannian and the affine flag variety.** For the facts we state here, we refer to [Fal03].

We set $\mathscr{K} := \mathbb{F}((z))$, $\mathscr{O} := \mathbb{F}[[z]]$, and consider the group ind-scheme $G_{\mathscr{K}}$ (denoted LG in [Fal03]) and its group subscheme $G_{\mathscr{O}}$ (denoted L^+G in [Fal03]).

We denote by $I^- \subset G_{\mathscr{O}}$ the Iwahori subgroup associated with B^-, i.e. the inverse image of B^- under the morphism $G_{\mathscr{O}} \to G$ sending z to 0. We consider the affine Grassmannian Gr and the affine flag variety Fl defined by

$$\mathrm{Gr} := G_{\mathscr{K}}/G_{\mathscr{O}}, \quad \mathrm{Fl} := G_{\mathscr{K}}/I^-.$$

We denote by $\pi : \mathrm{Fl} \to \mathrm{Gr}$ the projection morphism.

Any $\lambda \in \boldsymbol{X}^\vee$ defines a point $z^\lambda \in T_{\mathscr{K}} \subset G_{\mathscr{K}}$, hence a point $L_\lambda := z^\lambda G_{\mathscr{O}} \in \mathrm{Gr}$. We set

$$\mathrm{Gr}^\lambda := G_{\mathscr{O}} \cdot L_\lambda.$$

Then Gr^λ only depends on the W_{f}-orbit of λ. Moreover, the Bruhat decomposition implies that

$$\mathrm{Gr} = \bigsqcup_{\lambda \in \boldsymbol{X}^\vee_+} \mathrm{Gr}^\lambda.$$

We will denote by $j_\lambda : \mathrm{Gr}^\lambda \to \mathrm{Gr}$ the embedding.

For $\lambda \in \boldsymbol{X}^\vee_+$, we will denote by $P_\lambda \subset G$ the parabolic subgroup of G containing B^- associated with the subset of Δ_{s} consisting of those simple roots which are orthogonal to λ. Then P_λ is the stabilizer of L_λ in G, so that we have a canonical isomorphism $G/P_\lambda \xrightarrow{\sim} G \cdot L_\lambda$. Under this identification, it is known that the map $p_\lambda : \mathrm{Gr}^\lambda \to G/P_\lambda$ sending x to $\lim_{t \to 0} t \cdot x$ (where we consider the $\mathbb{G}_{\mathrm{m}}$-action on Gr via loop rotation) is a morphism of algebraic varieties, and realizes Gr^λ as an affine bundle over G/P_λ (see e.g. [NP01, Lem. 2.3]).

It is well known (see e.g. [Lus83] or [NP01, §2]) that if $\lambda \in \boldsymbol{X}^\vee_+$, then we have

$$\dim(\mathrm{Gr}^\lambda) = \langle \lambda, 2\rho \rangle = \sum_{\alpha \in \Delta^+} \langle \lambda, \alpha \rangle.$$

We denote by $\preccurlyeq$ the order on $\boldsymbol{X}^\vee_+$ determined by

$$\lambda \preccurlyeq \mu \quad \text{iff } \mu - \lambda \text{ is a sum of positive coroots.}$$

[1] This assumption holds in particular if G is semisimple of adjoint type.

Then for $\lambda, \mu \in \boldsymbol{X}^\vee_+$ we have

$$\mathrm{Gr}^\lambda \subset \overline{\mathrm{Gr}^\mu} \quad \text{iff} \quad \lambda \preccurlyeq \mu.$$

2.3. **Some categories of sheaves on** Gr **and** Fl**.** We let ℓ be a prime number which is different from p, and let $\Bbbk$ be either a finite extension of $\mathbb{Q}_\ell$, or the ring of integers in such an extension, or a finite field of characteristic ℓ. In this paper we will be concerned with the constructible derived categories $D^{\mathrm{b}}_c(\mathrm{Gr}, \Bbbk)$ and $D^{\mathrm{b}}_c(\mathrm{Fl}, \Bbbk)$ of étale $\Bbbk$-sheaves on Gr and Fl, respectively. If $K \subset G_{\mathscr{O}}$ is a subgroup, we will also denote by $D^{\mathrm{b}}_K(\mathrm{Gr}, \Bbbk)$ and $D^{\mathrm{b}}_K(\mathrm{Fl}, \Bbbk)$ the (constructible) K-equivariant derived category of $\Bbbk$-sheaves on Gr and Fl, in the sense of Bernstein–Lunts [BL94]. Each of these categories is endowed with the perverse t-structure, whose heart will be denoted $\mathrm{Perv}(\mathrm{Gr}, \Bbbk)$, $\mathrm{Perv}(\mathrm{Fl}, \Bbbk)$, $\mathrm{Perv}_K(\mathrm{Gr}, \Bbbk)$ and $\mathrm{Perv}_K(\mathrm{Fl}, \Bbbk)$ respectively.

Remark 2.1.
(1) Since Gr and Fl are ind-varieties and not varieties, the definition of the categories considered above requires some care; see e.g. [Nad05, §2.2] or [Gai01, App.] for details. We will not mention this point in the body of the paper, and simply refer to objects in these categories as complexes of sheaves.
(2) Recall that by [MV07] the category $\mathrm{Rep}(G^\vee_R)$ of algebraic representations of the group scheme $G^\vee_R$ over any Noetherian commutative base ring R of finite global dimension is equivalent to the corresponding category of spherical perverse sheaves on $\mathrm{Gr}_{\mathbb{C}}$ in its analytic topology. More restrictive assumptions on the base ring in the present paper come from our need to use the Artin–Schreier sheaf (see Section 3.2), which is only defined in the context of étale sheaves over a variety in positive characteristic; this setting yields categories of sheaves with coefficients as above. Notice however that some constructions involving the Artin–Schreier sheaf do have an analogue for constructible sheaves in the classical topology (see e.g. [Wan15] for the example of Fourier–Deligne transform). We expect that such a counterpart of the Whittaker category can also be defined (see [AG, Rem. 10.3.6] for a possible approach); this would allow one to extend our main result to more general coefficient rings.

If $K' \subset K \subset G_{\mathscr{O}}$ are subgroups, we will denote by

$$\mathsf{For}^K_{K'} : D^{\mathrm{b}}_K(\mathrm{Gr}, \Bbbk) \to D^{\mathrm{b}}_{K'}(\mathrm{Gr}, \Bbbk), \quad \mathsf{For}^K_{K'} : D^{\mathrm{b}}_K(\mathrm{Fl}, \Bbbk) \to D^{\mathrm{b}}_{K'}(\mathrm{Fl}, \Bbbk)$$

the natural forgetful functors. If K/K' is of finite type, these functors have both a right and a left adjoint, which will be denoted ${}^*\mathsf{Ind}^K_{K'}$ and ${}^!\mathsf{Ind}^K_{K'}$ respectively. If we write X for Gr or Fl, these functors can be described explicitly by

$${}^*\mathsf{Ind}^K_{K'}(\mathcal{F}) = a_*\big(\underline{\Bbbk} \mathbin{\widetilde{\boxtimes}} \mathcal{F}\big) \quad \text{and} \quad {}^!\mathsf{Ind}^K_{K'}(\mathcal{F}) = a_!\big(\underline{\Bbbk} \mathbin{\widetilde{\boxtimes}} \mathcal{F}\big)[2(\dim K/K')],$$

where $\underline{\Bbbk} \mathbin{\widetilde{\boxtimes}} (-)$ is the functor sending an object $\mathcal{F}$ to the only object in $D^{\mathrm{b}}_K(K \times^{K'} X, \Bbbk)$ whose pullback to $K \times X$ (an object of $D^{\mathrm{b}}_{K \times K'}(K \times X, \Bbbk)$, where K' acts on $K \times X$ via $h \cdot (g, x) = (gh^{-1}, h \cdot x)$) is isomorphic to $\underline{\Bbbk}_K \boxtimes^L_{\Bbbk} \mathcal{F}$. When $K' = \{1\}$ we will write For_K for $\mathsf{For}^K_{\{1\}}$.

2.4. **Convolution.** We will make extensive use of the convolution construction, defined as follows. Consider $\mathcal{F}, \mathcal{G}$ in $D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$, and the diagram

$$\mathrm{Gr} \times \mathrm{Gr} \xleftarrow{p^{\mathrm{Gr}}} G_{\mathscr{K}} \times \mathrm{Gr} \xrightarrow{q^{\mathrm{Gr}}} G_{\mathscr{K}} \times^{G_{\mathscr{O}}} \mathrm{Gr} \xrightarrow{m^{\mathrm{Gr}}} \mathrm{Gr},$$

where p^{Gr} and q^{Gr} are the quotient morphisms, and m^{Gr} is induced by the $G_{\mathscr{K}}$-action on Gr. Consider the action of $G_{\mathscr{O}} \times G_{\mathscr{O}}$ on $G_{\mathscr{K}} \times \mathrm{Gr}$ defined by

$$(g_1, g_2) \cdot (h_1, h_2 G_{\mathscr{O}}) = (g_1 h_1 (g_2)^{-1}, g_2 h_2 G_{\mathscr{O}}).$$

Then the functor $(q^{\mathrm{Gr}})^*$ induces an equivalence of categories

$$D^{\mathrm{b}}_{G_{\mathscr{O}}}(G_{\mathscr{K}} \times^{G_{\mathscr{O}}} \mathrm{Gr}, \Bbbk) \xrightarrow{\sim} D^{\mathrm{b}}_{G_{\mathscr{O}} \times G_{\mathscr{O}}}(G_{\mathscr{K}} \times \mathrm{Gr}, \Bbbk).$$

Hence there exists a unique object $\mathcal{F} \widetilde{\boxtimes} \mathcal{G}$ such that

$$(q^{\mathrm{Gr}})^*\big(\mathcal{F} \widetilde{\boxtimes} \mathcal{G}\big) = (p^{\mathrm{Gr}})^*\big(\mathcal{F} \overset{L}{\underset{\Bbbk}{\boxtimes}} \mathcal{G}\big).$$

The convolution product of $\mathcal{F}$ and $\mathcal{G}$ is defined by

$$\mathcal{F} \star^{G_{\mathscr{O}}} \mathcal{G} := (m^{\mathrm{Gr}})_*\big(\mathcal{F} \widetilde{\boxtimes} \mathcal{G}\big). \tag{2.1}$$

This construction endows the category $D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$ with the structure of a monoidal category. A similar formula defines a right action of this monoidal category on $D^{\mathrm{b}}_K(\mathrm{Gr}, \Bbbk)$, for any $K \subset G_{\mathscr{O}}$. (This action will again be denoted $\star^{G_{\mathscr{O}}}$.)

Remark 2.2. Note that if $\Bbbk$ is not a field, the convolution product considered above is *not* the same as the one considered (when $\mathcal{F}$ and $\mathcal{G}$ are perverse sheaves) in [MV07]: the product considered in [MV07] is rather defined as ${}^p\mathcal{H}^0(\mathcal{F} \star^{G_{\mathscr{O}}} \mathcal{G})$ in our notation.

Lemma 2.3. *Assume that $\Bbbk$ is a field. If $\mathcal{F}$ belongs to* $\mathrm{Perv}(\mathrm{Gr}, \Bbbk)$ *and $\mathcal{G}$ belongs to* $\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$*, then $\mathcal{F} \star^{G_{\mathscr{O}}} \mathcal{G}$ belongs to* $\mathrm{Perv}(\mathrm{Gr}, \Bbbk)$.

Proof. This claim follows from the description of convolution in terms of nearby cycles obtained in [Gai01, Prop. 6]. (In [Gai01], only the case of characteristic-0 coefficients is treated. However the same proof applies in general, simply replacing [Gai01, Prop. 1] by [MV07, Prop. 2.2].) □

Remark 2.4. The description of convolution in terms of nearby cycles as in [Gai01] works for general coefficients (if convolution is defined as in (2.1)). The nearby cycles functor is t-exact in this generality, but this description also involves a (derived) external tensor product. If $\Bbbk$ is not a field this tensor product operation is not t-exact, which explains the failure of Lemma 2.3 in this setting.

A very similar construction as the one considered above, based on the diagram

$$\mathrm{Fl} \times \mathrm{Fl} \xleftarrow{p^{\mathrm{Fl}}} G_{\mathscr{K}} \times \mathrm{Fl} \xrightarrow{q^{\mathrm{Fl}}} G_{\mathscr{K}} \times^{I^-} \mathrm{Fl} \xrightarrow{m^{\mathrm{Fl}}} \mathrm{Fl},$$

provides a convolution product $\star^{I^-}$ on $D^{\mathrm{b}}_{I^-}(\mathrm{Fl}, \Bbbk)$, which endows this category with the structure of a monoidal category, and defines a right action of this monoidal category on $D^{\mathrm{b}}_K(\mathrm{Fl}, \Bbbk)$, for any $K \subset G_{\mathscr{O}}$.

Again the same formulas, using the diagram

$$\mathrm{Fl} \times \mathrm{Gr} \xleftarrow{p^{\mathrm{Fl}}_{\mathrm{Gr}}} G_{\mathscr{K}} \times \mathrm{Gr} \xrightarrow{q^{\mathrm{Fl}}_{\mathrm{Gr}}} G_{\mathscr{K}} \times^{I^-} \mathrm{Gr} \xrightarrow{m^{\mathrm{Fl}}_{\mathrm{Gr}}} \mathrm{Gr},$$

allows to define a bifunctor

$$D^{\mathrm{b}}_K(\mathrm{Fl}, \Bbbk) \times D^{\mathrm{b}}_{I^-}(\mathrm{Gr}, \Bbbk) \to D^{\mathrm{b}}_K(\mathrm{Gr}, \Bbbk),$$

which will once again be denoted $\star^{I^-}$.

The following lemma is standard; its proof is left to interested readers.

Lemma 2.5. *For any subgroup $K \subset G_{\mathscr{O}}$, any $\mathcal{F}$ in $D^{\mathrm{b}}_K(\mathrm{Fl}, \Bbbk)$ and any $\mathcal{G}$ in $D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$, there exists a canonical isomorphism*

$$\mathcal{F} \star^{I^-} \mathsf{For}^{G_{\mathscr{O}}}_{I^-}(\mathcal{G}) \cong \pi_*(\mathcal{F}) \star^{G_{\mathscr{O}}} \mathcal{G}$$

in $D^{\mathrm{b}}_K(\mathrm{Gr}, \Bbbk)$.

In the following lemma we consider the convolution bifunctor

$$(-) \star^{G_{\mathscr{O}}} (-) : D^{\mathrm{b}}_K(\mathrm{Gr}, \Bbbk) \times D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Fl}, \Bbbk) \to D^{\mathrm{b}}_K(\mathrm{Fl}, \Bbbk)$$

(constructed once again using formulas similar to those above). Its proof is easy, and

left to the reader.

Lemma 2.6. *Let $\mathcal{F}$ in $D^{\mathrm{b}}_K(\mathrm{Gr}, \Bbbk)$ and $\mathcal{G}$ in $D^{\mathrm{b}}_{I^-}(\mathrm{Fl}, \Bbbk)$. Then there exists a canonical isomorphism*

$$\pi^*(\mathcal{F}) \star^{I^-} \mathcal{G} \cong \mathcal{F} \star^{G_{\mathscr{O}}} {}^*\mathsf{Ind}_{I^-}^{G_{\mathscr{O}}}(\mathcal{G})$$

in $D^{\mathrm{b}}_K(\mathrm{Fl}, \Bbbk)$.

3. Spherical vs. Iwahori–Whittaker

3.1. **Equivariant perverse sheaves on** Gr**.** For $\lambda \in \boldsymbol{X}^\vee_+$, we will denote by

$$\mathcal{J}_!(\lambda, \Bbbk) := {}^p\mathcal{H}^0\big((j_\lambda)_! \underline{\Bbbk}_{\mathrm{Gr}^\lambda}[\langle \lambda, 2\rho \rangle]\big), \quad \text{resp.} \quad \mathcal{J}_*(\lambda, \Bbbk) := {}^p\mathcal{H}^0\big((j_\lambda)_* \underline{\Bbbk}_{\mathrm{Gr}^\lambda}[\langle \lambda, 2\rho \rangle]\big),$$

the standard, resp. costandard, $G_{\mathscr{O}}$-equivariant perverse sheaf on Gr associated with λ.

We will also denote by $\mathcal{J}_{!*}(\lambda, \Bbbk)$ the image of any generator of the free rank-1 $\Bbbk$-module

$$\mathrm{Hom}_{\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)}(\mathcal{J}_!(\lambda, \Bbbk), \mathcal{J}_*(\lambda, \Bbbk)).$$

If $\Bbbk$ is a field then $\mathcal{J}_{!*}(\lambda, \Bbbk)$ is a simple perverse sheaf, which is both the head of $\mathcal{J}_!(\lambda, \Bbbk)$ and the socle of $\mathcal{J}_*(\lambda, \Bbbk)$.

Recall the notion of highest weight category, whose definition is spelled out e.g. in [Ric16, Def. 7.1]. (These conditions are obvious extensions of those considered in [BGS96, §3.2], which were preceded by a related study in [CPS88].)

Lemma 3.1. *Assume that $\Bbbk$ is a field. The category $\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$ is a highest weight category with weight poset $(\boldsymbol{X}^\vee_+, \preccurlyeq)$, standard objects $\{\mathcal{J}_!(\lambda, \Bbbk) : \lambda \in \boldsymbol{X}^\vee_+\}$, and costandard objects $\{\mathcal{J}_*(\lambda, \Bbbk) : \lambda \in \boldsymbol{X}^\vee_+\}$. Moreover, if $\mathrm{char}(\Bbbk) = 0$ then this category is semisimple.*

Proof. The first claim is an easy consequence of [MV07, Prop. 10.1(b)]; see [BR18, Prop. 1.12.4] for details. If $\mathrm{char}(\Bbbk) = 0$, the semisimplicity of the category $\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$ is well known: see [Gai01, Prop. 1] (or [BR18, §1.4] for an expanded version). □

Remark 3.2.

(1) If $\Bbbk$ is a field of characteristic 0, the semisimplicity of the category $\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$ implies in particular that the natural maps $\mathcal{J}_!(\lambda, \Bbbk) \to \mathcal{J}_{!*}(\lambda, \Bbbk) \to \mathcal{J}_*(\lambda, \Bbbk)$ are isomorphisms.

(2) For any coefficients $\Bbbk$, we have

$$\mathrm{Hom}_{D^{\mathrm{b}}\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)}\big(\mathcal{J}_!(\lambda, \Bbbk), \mathcal{J}_*(\mu, \Bbbk)[n]\big) = \begin{cases} \Bbbk & \text{if } n = 0 \text{ and } \lambda = \mu; \\ 0 & \text{otherwise.} \end{cases}$$

In fact, to prove this it suffices to prove the similar claim for perverse sheaves on Z, where $Z \subset \mathrm{Gr}$ is any closed finite union of $G_{\mathscr{O}}$-orbits containing Gr^λ and Gr^μ. In the case $\Bbbk$ is a field, this claim is a consequence of Lemma 3.1 (or rather its version for Z). The case when $\Bbbk$ is the ring of integers in a finite extension of $\mathbb{Q}_\ell$ follows. Indeed, since the category $\mathrm{Perv}_{G_{\mathscr{O}}}(Z, \Bbbk)$ has enough projectives we can consider the complex of $\Bbbk$-modules

$$M = R\,\mathrm{Hom}_{\mathrm{Perv}_{G_{\mathscr{O}}}(Z, \Bbbk)}(\mathcal{J}_!(\lambda, \Bbbk), \mathcal{J}_*(\mu, \Bbbk)).$$

If $\Bbbk_0$ is the residue field of $\Bbbk$, it is not difficult (using the results of [MV07, §8 & §10], and in particular the fact that $\Bbbk_0 \otimes^L_{\Bbbk} \mathcal{J}_?(\lambda, \Bbbk) \cong \mathcal{J}_?(\lambda, \Bbbk_0)$ for $? \in \{!, *\}$, see [MV07, Prop. 8.1]) to check that

$$\Bbbk_0 \overset{L}{\underset{\Bbbk}{\otimes}} M \cong R\operatorname{Hom}_{\operatorname{Perv}_{G_{\mathscr{O}}}(Z,\Bbbk_0)}(\mathcal{J}_!(\lambda, \Bbbk_0), \mathcal{J}_*(\mu, \Bbbk_0)).$$

We deduce that the left-hand side is isomorphic to $\Bbbk_0$ in the derived category of $\Bbbk_0$-vector spaces; this implies that M is isomorphic to $\Bbbk$ in the derived category of $\Bbbk$-modules.

In Section 4 we will also encounter some I^-_{u}-equivariant perverse sheaves on Gr, where I^-_{u} is the pro-unipotent radical of I^-. In particular, we have

$$\mathrm{Gr} = \bigsqcup_{\mu \in \boldsymbol{X}^\vee} I^-_{\mathrm{u}} \cdot L_\mu,$$

and we will denote by $\Delta^{\mathrm{Gr}}_\mu(\Bbbk)$, resp. $\nabla^{\mathrm{Gr}}_\mu(\Bbbk)$, the standard, resp. costandard, perverse sheaf associated with μ, i.e. the !-direct image (resp. $*$-direct image) of the constant perverse sheaf of rank 1 on $I^-_{\mathrm{u}} \cdot L_\mu$. (These objects are perverse sheaves thanks to [BBDG82, Cor. 4.1.3], because the orbits $I^-_{\mathrm{u}} \cdot L_\mu$ are affine spaces.)

3.2. **Category of Iwahori–Whittaker perverse sheaves.** We now denote by $I^+ \subset G_{\mathscr{O}}$ the Iwahori subgroup associated with B^+. We also denote by I^+_{u} the pro-unipotent radical of I^+, i.e. the inverse image of U^+ under the map $I^+ \to B^+$.

We assume that there exists a primitive p-th root of unity in $\Bbbk$, and fix one. This choice determines a character ψ of the prime subfield of $\mathbb{F}$ (with values in $\Bbbk^\times$), and we denote by $\mathcal{L}^{\Bbbk}_\psi$ the corresponding Artin–Schreier local system on $\mathbb{G}_{\mathrm{a}}$. (Below, some arguments using Verdier duality will also involve the Artin–Schreier local system $\mathcal{L}^{\Bbbk}_{-\psi}$ associated with the character ψ^{-1}; clearly these two versions play similar roles.) We also consider the "generic" character $\chi : U^+ \to \mathbb{G}_{\mathrm{a}}$ defined as the composition

$$U^+ \twoheadrightarrow U^+/[U^+, U^+] \xleftarrow[\sim]{\prod_\alpha u_\alpha} \prod_{\alpha \in \Delta_{\mathrm{s}}} \mathbb{G}_{\mathrm{a}} \xrightarrow{+} \mathbb{G}_{\mathrm{a}},$$

and denote by χ_{I^+} its composition with the projection $I^+_{\mathrm{u}} \twoheadrightarrow U^+$. We can then define the "Iwahori–Whittaker" derived category

$$D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$$

as the $(I^+_{\mathrm{u}}, \chi^*_{I^+}(\mathcal{L}^{\Bbbk}_\psi))$-equivariant constructible derived category of $\Bbbk$-sheaves on Gr (see e.g. [AR16, App. A] for a review of the construction of this category). This category admits a perverse t-structure, whose heart will be denoted $\operatorname{Perv}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$, and moreover the "realization functor"

$$D^{\mathrm{b}}\operatorname{Perv}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk) \to D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$$

is an equivalence of triangulated categories.

Note that any ring of coefficients considered above appears in an ℓ-modular triple $(\mathbb{K}, \mathbb{O}, \mathbb{L})$ where $\mathbb{K}$ is a finite extension of $\mathbb{Q}_\ell$, $\mathbb{O}$ is its ring of integers, and $\mathbb{L}$ is the residue field of $\mathbb{O}$. In this setting the embedding $\mathbb{O} \hookrightarrow \mathbb{K}$ and the projection $\mathbb{O} \to \mathbb{L}$ induce bijections between the p-th roots of unity in $\mathbb{O}$, $\mathbb{K}$ and $\mathbb{L}$. Therefore, choosing a primitive root in any of these rings provides primitive roots in all three rings, and we can then consider extension of scalars functors

$$\mathbb{K} \overset{L}{\underset{\mathbb{O}}{\otimes}}(-) : D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \mathbb{O}) \to D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \mathbb{K}), \qquad \mathbb{L} \overset{L}{\underset{\mathbb{O}}{\otimes}}(-) : D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \mathbb{O}) \to D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \mathbb{L}).$$

These functors will play a crucial role in our arguments below. (Here the first functor is t-exact, but the second one is only right t-exact.) For $\lambda \in \boldsymbol{X}^\vee$ we set

$$X_\lambda := I^+ \cdot L_\lambda.$$

Then again we have

$$\mathrm{Gr} = \bigsqcup_{\lambda \in \boldsymbol{X}^\vee} X_\lambda.$$

Lemma 3.3. *The orbit X_λ supports an $(I^+_{\mathrm{u}}, \chi^*_{I^+}(\mathcal{L}^{\Bbbk}_\psi))$-equivariant local system iff $\lambda \in \boldsymbol{X}^\vee_{++}$.*

Sketch of proof. Let $\lambda \in \boldsymbol{X}^\vee_+$, and consider the affine bundle $\mathrm{Gr}^\lambda \to G/P_\lambda$ (see Section 2.2). The decomposition of Gr^λ in I^+_{u}-orbits is obtained by pullback from the decomposition of G/P_λ into U^+-orbits; in particular, for $\mu \in W_{\mathrm{f}} \cdot \lambda$, X_μ supports an $(I^+_{\mathrm{u}}, \chi^*_{I^+}(\mathcal{L}^{\Bbbk}_\psi))$-equivariant local system iff its image in G/P_λ is a free U^+-orbit. If $\lambda \notin \boldsymbol{X}^\vee_{++}$ there is no such orbit in G/P_λ, and if $\lambda \in \boldsymbol{X}^\vee_{++}$ there is exactly one, corresponding to X_λ. □

Note that if $\lambda \in \boldsymbol{X}^\vee_{++}$, since X_λ is open dense in Gr^λ we have

$$\dim(X_\lambda) = \langle \lambda, 2\rho \rangle \tag{3.1}$$

and

$$X_\lambda \subset \overline{X_\mu} \quad \text{iff} \quad \lambda \preccurlyeq \mu.$$

For $\lambda \in \boldsymbol{X}^\vee_{++}$ we will denote by

$$\Delta^{\mathcal{IW}}_\lambda(\Bbbk), \quad \text{resp.} \quad \nabla^{\mathcal{IW}}_\lambda(\Bbbk),$$

the standard, resp. costandard, $(I^+_{\mathrm{u}}, \chi^*_{I^+}(\mathcal{L}^{\Bbbk}_\psi))$-equivariant perverse sheaf on Gr associated with λ, i.e. the !-extension, resp. $*$-extension, to Gr of the free rank-1 $(I^+_{\mathrm{u}}, \chi^*_{I^+}(\mathcal{L}^{\Bbbk}_\psi))$-equivariant perverse sheaf on X_λ. (Once again, these objects are perverse sheaves thanks to [BBDG82, Cor. 4.1.3].) We will also denote by $\mathrm{IC}^{\mathcal{IW}}_\lambda(\Bbbk)$ the image of any generator of the rank-1 free $\Bbbk$-module

$$\mathrm{Hom}_{\mathrm{Perv}_{\mathcal{IW}}(\mathrm{Gr},\Bbbk)}(\Delta^{\mathcal{IW}}_\lambda(\Bbbk), \nabla^{\mathcal{IW}}_\lambda(\Bbbk)).$$

If $\Bbbk$ is a field then $\mathrm{IC}^{\mathcal{IW}}_\lambda(\Bbbk)$ is a simple perverse sheaf.

Note that since ς is minimal in $\boldsymbol{X}^\vee_{++}$ for $\preccurlyeq$, we have

$$\Delta^{\mathcal{IW}}_\varsigma(\Bbbk) = \nabla^{\mathcal{IW}}_\varsigma(\Bbbk) = \mathrm{IC}^{\mathcal{IW}}_\varsigma(\Bbbk). \tag{3.2}$$

Lemma 3.4. *Assume that $\Bbbk$ is a field of characteristic 0. Then the i-th cohomology of the stalks of $\mathrm{IC}^{\mathcal{IW}}_\lambda(\Bbbk)$ vanish unless $i \equiv \dim X_\lambda \pmod 2$.*

Sketch of proof. Since the morphism π is smooth, by standard properties of perverse sheaves (see e.g. [BBDG82, §4.2.6]) it suffices to prove a similar statement on Fl instead of Gr. Now the Decomposition Theorem implies that all simple $(I^+_{\mathrm{u}}, \chi^*_{I^+}(\mathcal{L}^{\Bbbk}_\psi))$-equivariant perverse sheaves on Fl can be obtained from the one corresponding to the orbit of the base point by convolving on the right with I^--equivariant simple perverse sheaves on Fl corresponding to orbits of dimension either 0 or 1. Standard arguments (going back at least to [Spr82]) show that these operations preserve the parity-vanishing property of stalks, and the claim follows. □

Remark 3.5. Assume that $\Bbbk$ is a field. Following [JMW14][2] we will say that an object of $D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$ is *even*, resp. *odd*, if its restriction and corestriction to each

[2]In [JMW14] the authors consider the setting of "ordinary" constructible complexes. However, as observed already in [RW18, §11.1] or [AMRW19, §6.2], their considerations apply verbatim in our Iwahori–Whittaker setting.

stratum is concentrated in even, resp. odd, degrees, and that it is *parity* if it is isomorphic to a direct sum $\mathcal{F} \oplus \mathcal{F}'$ with $\mathcal{F}$ even and $\mathcal{F}'$ odd. Using this language, Lemma 3.4 states that if $\mathrm{char}(\Bbbk) = 0$ then the objects $\mathrm{IC}_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk)$ are parity, of the same parity as $\dim(X_\lambda)$.

Corollary 3.6. *Assume that $\Bbbk$ is a field. The category $\mathrm{Perv}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr}, \Bbbk)$ is a highest weight category with weight poset $(\boldsymbol{X}^\vee_{++}, \preccurlyeq)$, standard objects $\{\Delta_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk) : \lambda \in \boldsymbol{X}^\vee_{++}\}$, and costandard objects $\{\nabla_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk) : \lambda \in \boldsymbol{X}^\vee_{++}\}$. Moreover, if $\mathrm{char}(\Bbbk) = 0$ then this category is semisimple.*

Proof. The first claim is standard, as e.g. in [BGS96, §3.3]. For the second claim, we observe that the orbits X_λ (for $\lambda \in \boldsymbol{X}^\vee_{++}$) have dimensions of constant parity on each connected component of Gr, see (3.1).
Using this and Lemma 3.4, the semisimplicity can be proved exactly as in the case of the category $\mathrm{Perv}_{G_\mathscr{O}}(\mathrm{Gr}, \Bbbk)$. Namely, we have to prove that

$$\mathrm{Ext}^1_{\mathrm{Perv}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr},\Bbbk)}(\mathrm{IC}_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk), \mathrm{IC}_\mu^{\mathcal{I}\mathcal{W}}(\Bbbk)) = \mathrm{Hom}_{D^{\mathrm{b}}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr},\Bbbk)}(\mathrm{IC}_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk), \mathrm{IC}_\mu^{\mathcal{I}\mathcal{W}}(\Bbbk)[1])$$

vanishes for any λ, μ. If X_λ and X_μ belong to different connected components of Gr then this claim is obvious; otherwise $\mathrm{IC}_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk)$ and $\mathrm{IC}_\mu^{\mathcal{I}\mathcal{W}}(\Bbbk)$ are either both even or both odd (see Remark 3.5), so that the desired vanishing follows from [JMW14, Cor. 2.8]. □

Remark 3.7. (1) Once Corollary 3.6 is known, one can refine Lemma 3.4 drastically: if $\Bbbk$ is a field of characteristic 0, then the simple perverse sheaves $\mathrm{IC}_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk)$ are clean, in the sense that if $i_\mu : X_\mu \to \mathrm{Gr}$ is the embedding, for any $\mu \neq \lambda$ we have

$$i_\mu^*\big(\mathrm{IC}_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk)\big) = i_\mu^!\big(\mathrm{IC}_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk)\big) = 0. \tag{3.3}$$

In fact, as in Remark 3.2(1), the semisimplicity claim in Corollary 3.6 implies that the natural maps

$$\Delta_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk) \to \mathrm{IC}_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk) \to \nabla_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk)$$

are isomorphisms, which is equivalent to (3.3). (See also [ABB+05, Cor. 2.2.3] for a different proof of (3.3).) This observation can be used to give a new proof of the main result of [FGV01], hence of the geometric Casselman–Shalika formula.

(2) The same arguments as in Remark 3.2(2) show that for any coefficients $\Bbbk$, any $\lambda, \mu \in \boldsymbol{X}^\vee_{++}$ and any $n \in \mathbb{Z}$ we have

$$\mathrm{Hom}_{D^{\mathrm{b}}\mathrm{Perv}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr},\Bbbk)}\left(\Delta_\lambda^{\mathcal{I}\mathcal{W}}(\Bbbk), \nabla_\mu^{\mathcal{I}\mathcal{W}}(\Bbbk)[n]\right) = \begin{cases} \Bbbk & \text{if } \lambda = \mu \text{ and } n = 0; \\ 0 & \text{otherwise.} \end{cases}$$

(In this case, the existence of enough projectives in $\mathrm{Perv}_{\mathcal{I}\mathcal{W}}(Z, \Bbbk)$ can be checked using the techniques of [RSW14, §2].)

3.3. **Statement.** We consider the functor

$$\Phi : D^{\mathrm{b}}_{G_\mathscr{O}}(\mathrm{Gr}, \Bbbk) \to D^{\mathrm{b}}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr}, \Bbbk)$$

defined by

$$\Phi(\mathcal{F}) = \Delta_\varsigma^{\mathcal{I}\mathcal{W}}(\Bbbk) \star^{G_\mathscr{O}} \mathcal{F}.$$

In view of (3.2) (or, alternatively, arguing as in [BBM04]), in this definition $\Delta_\varsigma^{\mathcal{I}\mathcal{W}}(\Bbbk)$ can be replaced by $\nabla_\varsigma^{\mathcal{I}\mathcal{W}}(\Bbbk)$ or $\mathrm{IC}_\varsigma^{\mathcal{I}\mathcal{W}}(\Bbbk)$; in particular this shows that the conjugate of Φ by Verdier duality is the similar functor using the character ψ^{-1} instead of ψ.

Lemma 3.8. *The functor Φ is t-exact for the perverse t-structures.*

Proof. In the case where $\Bbbk$ is a field, the claim follows from Lemma 2.3. The general case follows using an ℓ-modular triple $(\mathbb{K}, \mathbb{O}, \mathbb{L})$ as in Section 3.2 with $\Bbbk = \mathbb{O}$, and the associated extension of scalars functors. Namely, if $\mathcal{F}$ is in $\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \mathbb{O})$, then

$$\mathbb{K} \overset{L}{\underset{\mathbb{O}}{\otimes}} \Phi(\mathcal{F}) \cong \Phi(\mathbb{K} \overset{L}{\underset{\mathbb{O}}{\otimes}} \mathcal{F})$$

is perverse; hence any perverse cohomology object ${}^p\mathcal{H}^i(\Phi(\mathcal{F}))$ with $i \neq 0$ is torsion. On the other hand,

$$\mathbb{L} \overset{L}{\underset{\mathbb{O}}{\otimes}} \Phi(\mathcal{F}) \cong \Phi(\mathbb{L} \overset{L}{\underset{\mathbb{O}}{\otimes}} \mathcal{F})$$

lives in perverse degrees -1 and 0 since $\mathbb{L} \otimes^L_{\mathbb{O}} \mathcal{F}$ lives in these degrees. If ${}^p\mathcal{H}^i(\Phi(\mathcal{F}))$ were nonzero for some $i > 0$, then taking i maximal with this property we would obtain that ${}^p\mathcal{H}^i(\mathbb{L} \otimes^L_{\mathbb{O}} \Phi(\mathcal{F})) \neq 0$, a contradiction. On the other hand, if ${}^p\mathcal{H}^i(\Phi(\mathcal{F}))$ was nonzero for some $i < 0$, then taking i minimal with this property we would obtain that ${}^p\mathcal{H}^{i-1}(\mathbb{L} \otimes^L_{\mathbb{O}} \Phi(\mathcal{F})) \neq 0$, a contradiction again. □

We will denote by

$$\Phi^0 : \mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk) \to \mathrm{Perv}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr}, \Bbbk)$$

the restriction of Φ to the hearts of the perverse t-structures, so that Φ^0 is an exact functor between abelian categories. The main result of this section is the following theorem, whose proof will be given in the next subsection.

Theorem 3.9. *The functor Φ^0 is an equivalence of categories.*

3.4. **Proof of Theorem 3.9.** As explained in Section 3.2, any ring of coefficients considered above appears in an ℓ-modular triple $(\mathbb{K}, \mathbb{O}, \mathbb{L})$ where $\mathbb{K}$ is a finite extension of $\mathbb{Q}_\ell$, $\mathbb{O}$ is its ring of integers, and $\mathbb{L}$ is the residue field of $\mathbb{O}$. Therefore we fix such a triple, and will treat the three cases in parallel.

The starting point of our proof will be the *geometric Casselman–Shalika formula*, first conjectured in [FGKV98] and then proved independently in [FGV01] and [NP01] (see also Remark 3.7(1)). We consider the composition

$$\chi_{U^+_{\mathscr{K}}} : U^+_{\mathscr{K}} \xrightarrow{\chi_{\mathscr{K}}} (\mathbb{G}_{\mathrm{a}})_{\mathscr{K}} \to \mathbb{G}_{\mathrm{a}},$$

where the second map is the "residue" morphism defined by

$$\sum_{i \in \mathbb{Z}} f_i z^i \mapsto f_{-1}.$$

For $\mu \in \boldsymbol{X}^\vee_+$ we set $S_\mu := U^+_{\mathscr{K}} \cdot L_\mu$; then there exists a unique function $\chi_\mu : S_\mu \to \mathbb{G}_{\mathrm{a}}$ such that $\chi_\mu(u \cdot L_\mu) = \chi_{U^+_{\mathscr{K}}}(u)$ for any $u \in U^+_{\mathscr{K}}$. The geometric Casselman–Shalika formula states that for $\lambda, \mu \in \boldsymbol{X}^\vee_+$ we have

$$\mathsf{H}^i_c\big(S_\mu, \jmath_{!*}(\lambda, \mathbb{K})_{|S_\mu} \otimes_{\mathbb{K}} \chi_\mu^*(\mathcal{L}^{\mathbb{K}}_\psi)\big) = \begin{cases} \mathbb{K} & \text{if } \lambda = \mu \text{ and } i = \langle 2\rho, \lambda \rangle; \\ 0 & \text{otherwise.} \end{cases} \tag{3.4}$$

In the following lemma, we denote by $\chi'_\mu : z^{-\varsigma} X_{\mu+\varsigma} \to \mathbb{G}_{\mathrm{a}}$ the unique function such that $\chi'_\mu(z^{-\varsigma} \cdot u \cdot L_{\mu+\varsigma}) = \chi_{I^+}(u)$ for $u \in I^+_{\mathrm{u}}$.

Lemma 3.10. *For $\Bbbk \in \{\mathbb{K}, \mathbb{O}, \mathbb{L}\}$, for any $\lambda, \mu \in \boldsymbol{X}^\vee_+$ with $\lambda \neq \mu$ we have*

$$\mathsf{H}^{\langle \lambda+\mu, 2\rho \rangle}_c\big(\mathrm{Gr}^\lambda \cap (z^{-\varsigma} X_{\mu+\varsigma}), (\chi'_\mu)^*(\mathcal{L}^{\Bbbk}_\psi)_{|\mathrm{Gr}^\lambda \cap (z^{-\varsigma} X_{\mu+\varsigma})}\big) = 0.$$

Proof. For $\alpha \in \Delta$ and $n \in \mathbb{Z}_{\geqslant 0}$ we denote by $U_{\alpha,n} \subset G_{\mathscr{O}}$ the image of the morphism $x \mapsto u_\alpha(xz^n)$. As explained e.g. in [NP01, Lem. 2.2], the action on $L_{\mu+\varsigma}$ induces an isomorphism

$$\prod_{\alpha\in\Delta^+} \prod_{j=0}^{\langle \mu+\varsigma,\alpha\rangle-1} U_{\alpha,j} \xrightarrow{\sim} X_{\mu+\varsigma}.$$

Multiplying by $z^{-\varsigma}$ we deduce that $z^{-\varsigma}X_{\mu+\varsigma} \subset S_\mu$, and moreover that χ'_μ is the restriction of χ_μ to $z^{-\varsigma}X_{\mu+\varsigma}$. By [MV07, Th. 3.2], we have $\dim(\mathrm{Gr}^\lambda \cap S_\mu) = \langle \lambda+\mu, \rho\rangle$; it follows that $\dim(\mathrm{Gr}^\lambda \cap (z^{-\varsigma}X_{\mu+\varsigma})) \leqslant \langle \lambda+\mu,\rho\rangle$. If this inequality is strict, then our vanishing claim is obvious (see e.g. [FK88, Th. I.8.8]). Otherwise, each irreducible component of $\mathrm{Gr}^\lambda \cap (z^{-\varsigma}X_{\mu+\varsigma})$ of dimension $\langle \lambda+\mu,\rho\rangle$ is dense (hence open) in an irreducible component of $\mathrm{Gr}^\lambda \cap S_\mu$; therefore to prove the lemma it suffices to prove that

$$\mathsf{H}_c^{\langle \lambda+\mu,2\rho\rangle}\big(\mathrm{Gr}^\lambda \cap S_\mu, \chi_\mu^*(\mathcal{L}_\psi^{\Bbbk})_{|\mathrm{Gr}^\lambda\cap S_\mu}\big) = 0. \tag{3.5}$$

Finally, we note that since we are considering the top cohomology, the $\mathbb{O}$-module $\mathsf{H}_c^{\langle \lambda+\mu,2\rho\rangle}\big(\mathrm{Gr}^\lambda \cap S_\mu, \chi_\mu^*(\mathcal{L}_\psi^{\mathbb{O}})_{|\mathrm{Gr}^\lambda\cap S_\mu}\big)$ is free, and the natural morphisms

$$\mathbb{K} \otimes_{\mathbb{O}} \mathsf{H}_c^{\langle \lambda+\mu,2\rho\rangle}\big(\mathrm{Gr}^\lambda \cap S_\mu, \chi_\mu^*(\mathcal{L}_\psi^{\mathbb{O}})_{|\mathrm{Gr}^\lambda\cap S_\mu}\big) \to \mathsf{H}_c^{\langle \lambda+\mu,2\rho\rangle}\big(\mathrm{Gr}^\lambda \cap S_\mu, \chi_\mu^*(\mathcal{L}_\psi^{\mathbb{K}})_{|\mathrm{Gr}^\lambda\cap S_\mu}\big),$$
$$\mathbb{L} \otimes_{\mathbb{O}} \mathsf{H}_c^{\langle \lambda+\mu,2\rho\rangle}\big(\mathrm{Gr}^\lambda \cap S_\mu, \chi_\mu^*(\mathcal{L}_\psi^{\mathbb{O}})_{|\mathrm{Gr}^\lambda\cap S_\mu}\big) \to \mathsf{H}_c^{\langle \lambda+\mu,2\rho\rangle}\big(\mathrm{Gr}^\lambda \cap S_\mu, \chi_\mu^*(\mathcal{L}_\psi^{\mathbb{L}})_{|\mathrm{Gr}^\lambda\cap S_\mu}\big)$$

are isomorphisms; hence it suffices to prove (3.5) in case $\Bbbk = \mathbb{K}$.
So, from now on we assume that $\Bbbk = \mathbb{K}$. The geometric Casselman–Shalika formula (3.4) implies that for any $\mathcal{F}$ in $\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \mathbb{K})$ we have $\mathsf{H}_c^i(S_\mu, \mathcal{F}_{|S_\mu} \otimes_{\mathbb{K}} \chi_\mu^*(\mathcal{L}_\psi^{\mathbb{K}})) = 0$ for $i \neq \langle 2\rho, \mu\rangle$; therefore the morphism $(j_\lambda)_! \underline{\mathbb{K}}_{\mathrm{Gr}^\lambda}[\langle \lambda, 2\rho\rangle] \to \mathcal{J}_!(\lambda, \mathbb{K})$ induces an isomorphism

$$\mathsf{H}_c^{\langle \lambda+\mu,2\rho\rangle}\Big(S_\mu, \big((j_\lambda)_!\underline{\mathbb{K}}_{\mathrm{Gr}^\lambda}\big)_{|S_\mu} \otimes_{\mathbb{K}} \chi_\mu^*(\mathcal{L}_\psi^{\mathbb{K}})\Big) \xrightarrow{\sim} \mathsf{H}_c^{\langle \mu,2\rho\rangle}\big(S_\mu, \mathcal{J}_!(\lambda,\mathbb{K}) \otimes_{\mathbb{K}} \chi_\mu^*(\mathcal{L}_\psi^{\mathbb{K}})\big).$$

Now we have $\mathcal{J}_!(\lambda,\mathbb{K}) \cong \mathcal{J}_{!*}(\lambda,\mathbb{K})$ by Remark 3.2(1); hence the right-hand side vanishes if $\lambda \neq \mu$ by (3.4). On the other hand, the base change theorem shows that the left-hand side identifies with $\mathsf{H}_c^{\langle \lambda+\mu,2\rho\rangle}\big(\mathrm{Gr}^\lambda \cap S_\mu, \chi_\mu^*(\mathcal{L}_\psi^{\mathbb{K}})_{|\mathrm{Gr}^\lambda\cap S_\mu}\big)$; we have therefore proved (3.5) in this case, hence the lemma. □

Proposition 3.11. *For $\Bbbk \in \{\mathbb{K}, \mathbb{L}\}$, for any $\lambda, \mu \in \boldsymbol{X}_+^\vee$ with $\lambda \neq \mu$ we have*

$$\mathrm{Hom}_{\mathrm{Perv}_{\mathcal{J}\mathcal{W}}(\mathrm{Gr},\Bbbk)}\big(\Phi^0(\mathcal{J}_!(\lambda,\Bbbk)), \nabla^{\mathcal{J}\mathcal{W}}_{\mu+\varsigma}(\Bbbk)\big) = 0.$$

Proof. First, by exactness of Φ we see that the morphism $(j_\lambda)_!\underline{\Bbbk}_{\mathrm{Gr}^\lambda}[\langle \lambda, 2\rho\rangle] \to \mathcal{J}_!(\lambda,\Bbbk)$ induces an isomorphism

$$\mathrm{Hom}_{D^{\mathrm{b}}_{\mathcal{J}\mathcal{W}}(\mathrm{Gr},\Bbbk)}\big(\Phi((j_\lambda)_!\underline{\Bbbk}_{\mathrm{Gr}^\lambda}[\langle \lambda,2\rho\rangle]), \nabla^{\mathcal{J}\mathcal{W}}_{\mu+\varsigma}(\Bbbk)\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Perv}_{\mathcal{J}\mathcal{W}}(\mathrm{Gr},\Bbbk)}\big(\Phi^0(\mathcal{J}_!(\lambda,\Bbbk)), \nabla^{\mathcal{J}\mathcal{W}}_{\mu+\varsigma}(\Bbbk)\big).$$

Let J be the stabilizer of the point L_ς in I_{u}^+. Then $\chi_{I^+}^*(\mathcal{L}_\psi^{\Bbbk})$ is trivial on J, so that we have a forgetful functor

$$D^{\mathrm{b}}_{\mathcal{J}\mathcal{W}}(\mathrm{Gr},\Bbbk) \to D^{\mathrm{b}}_J(\mathrm{Gr},\Bbbk).$$

By the same considerations as in Section 2.3, this functor admits a left adjoint, denoted ${}^!\mathsf{Ind}_J^{(I_{\mathrm{u}}^+,\chi_{I^+})}$. Moreover, since J is pro-unipotent the forgetful functor $D^{\mathrm{b}}_J(\mathrm{Gr},\Bbbk) \to D^{\mathrm{b}}_c(\mathrm{Gr},\Bbbk)$ is fully faithful.
If we denote by F_ς the direct image under the automorphism $x \mapsto z^\varsigma \cdot x$ of Gr, then from the definition we see that

$$\Phi(\mathcal{F}) = {}^!\mathsf{Ind}_J^{(I_{\mathrm{u}}^+,\chi_{I^+})} \circ \mathsf{F}_\varsigma \circ \mathsf{For}^{G_{\mathscr{O}}}_{z^{-\varsigma}Jz^\varsigma}(\mathcal{F})[-\langle \varsigma, 2\rho\rangle] \tag{3.6}$$

for any $\mathcal{F}$ in $D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Gr},\Bbbk)$. In the setting of the proposition, we deduce an isomorphism

$$\mathrm{Hom}_{D^{\mathrm{b}}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr},\Bbbk)}\big(\Phi((j_\lambda)_!\underline{\Bbbk}_{\mathrm{Gr}^\lambda}[\langle\lambda,2\rho\rangle]),\nabla^{\mathcal{I}\mathcal{W}}_{\mu+\varsigma}(\Bbbk)\big)\cong\mathrm{Hom}_{D^{\mathrm{b}}_c(\mathrm{Gr},\Bbbk)}\big((j_\lambda)_!\underline{\Bbbk}_{\mathrm{Gr}^\lambda}[\langle\lambda,2\rho\rangle],\mathsf{F}_\varsigma^{-1}(\nabla^{\mathcal{I}\mathcal{W}}_{\mu+\varsigma}(\Bbbk))[\langle\varsigma,2\rho\rangle]\big).$$

Now $\mathsf{F}_\varsigma^{-1}(\nabla^{\mathcal{I}\mathcal{W}}_{\mu+\varsigma}(\Bbbk))$ identifies with the $*$-pushforward of $(\chi'_\mu)^*(\mathcal{L}^\Bbbk_\psi)[\langle\mu+\varsigma,2\rho\rangle]$ under the embedding $z^{-\varsigma}X_{\mu+\varsigma}\to\mathrm{Gr}$. Hence, by the base change theorem, the right-hand side identifies with

$$\mathsf{H}^{\langle\mu+2\varsigma-\lambda,2\rho\rangle}(\mathrm{Gr}^\lambda\cap z^{-\varsigma}X_{\mu+\varsigma},a^!(\chi'_\mu)^*(\mathcal{L}^\Bbbk_\psi)),$$

where $a:\mathrm{Gr}^\lambda\cap z^{-\varsigma}X_{\mu+\varsigma}\hookrightarrow z^{-\varsigma}X_{\mu+\varsigma}$ is the embedding.
So, we now need to show that $\mathsf{H}^{\langle\mu+2\varsigma-\lambda,2\rho\rangle}(\mathrm{Gr}^\lambda\cap z^{-\varsigma}X_{\mu+\varsigma},a^!(\chi'_\mu)^*(\mathcal{L}^\Bbbk_\psi))$ vanishes. If b denotes the unique map $z^{-\varsigma}X_{\mu+\varsigma}\to\mathrm{pt}$, then by Verdier duality we have

$$\begin{aligned}&\mathsf{H}^{\langle\mu+2\varsigma-\lambda,2\rho\rangle}(\mathrm{Gr}^\lambda\cap z^{-\varsigma}X_{\mu+\varsigma},a^!(\chi'_\mu)^*(\mathcal{L}^\Bbbk_\psi))^*=\mathsf{H}^{\langle\mu+2\varsigma-\lambda,2\rho\rangle}(b_*a^!(\chi'_\mu)^*(\mathcal{L}^\Bbbk_\psi))^*\\&\cong\mathsf{H}^{\langle\lambda-2\varsigma-\mu,2\rho\rangle}(b_!a^*\mathbb{D}_{z^{-\varsigma}X_{\mu+\varsigma}}((\chi'_\mu)^*(\mathcal{L}^\Bbbk_\psi))).\end{aligned}$$

Now since $z^{-\varsigma}X_{\mu+\varsigma}$ is smooth of dimension $\langle\mu+\varsigma,2\rho\rangle$ we have an isomorphism $\mathbb{D}_{z^{-\varsigma}X_{\mu+\varsigma}}((\chi'_\mu)^*(\mathcal{L}^\Bbbk_\psi))\cong(\chi'_\mu)^*(\mathcal{L}^\Bbbk_{-\psi})[2\langle\mu+\varsigma,2\rho\rangle]$, which shows that

$$\mathsf{H}^{\langle\mu+2\varsigma-\lambda,2\rho\rangle}(\mathrm{Gr}^\lambda\cap z^{-\varsigma}X_{\mu+\varsigma},a^!(\chi'_\mu)^*(\mathcal{L}^\Bbbk_\psi))^*\cong\mathsf{H}^{\langle\mu+\lambda,2\rho\rangle}_c(\mathrm{Gr}^\lambda\cap z^{-\varsigma}X_{\mu+\varsigma},a^*(\chi'_\mu)^*(\mathcal{L}^\Bbbk_{-\psi})).$$

The right-hand side vanishes by Lemma 3.10, hence so does the left-hand side, which completes the proof. □

We can finally give the proof of Theorem 3.9.

Proof of Theorem 3.9. Let $\tau:G_{\mathscr{K}}\to\mathrm{Gr}$ be the projection. Let $\lambda\in\boldsymbol{X}^\vee_+$, and denote by $m_{\varsigma,\lambda}$ the restriction of m^{Gr} to $\tau^{-1}(\overline{\mathrm{Gr}^\varsigma})\times^{G_{\mathscr{O}}}\overline{\mathrm{Gr}^\lambda}$ (see Section 2.4 for the notation). Then it is well known that:

- $m_{\varsigma,\lambda}$ takes values in $\overline{\mathrm{Gr}^{\lambda+\varsigma}}=\overline{X_{\lambda+\varsigma}}$;
- its restriction to the preimage of $X_{\lambda+\varsigma}$ is an isomorphism;
- this preimage is contained in $\tau^{-1}(X_\varsigma)\times^{G_{\mathscr{O}}}\mathrm{Gr}^\lambda$.

These properties imply that the perverse sheaf $\Phi^0(\mathcal{J}_!(\lambda,\Bbbk))$ is supported on $\overline{X_{\lambda+\varsigma}}$, and that its restriction to $X_{\lambda+\varsigma}$ is a perversely shifted local system of rank 1. The same comments apply to $\Phi^0(\mathcal{J}_*(\lambda,\Bbbk))$. Hence there exist canonical morphisms

$$f^\Bbbk_\lambda:\Delta^{\mathcal{I}\mathcal{W}}_{\lambda+\varsigma}(\Bbbk)\to\Phi^0(\mathcal{J}_!(\lambda,\Bbbk))\quad\text{and}\quad g^\Bbbk_\lambda:\Phi^0(\mathcal{J}_*(\lambda,\Bbbk))\to\nabla^{\mathcal{I}\mathcal{W}}_{\lambda+\varsigma}(\Bbbk)$$

whose restrictions to $X_{\lambda+\varsigma}$ are isomorphisms.
We claim that $f^\Bbbk_\lambda$ is an isomorphism. First we note that all of our constructions are compatible with extension-of-scalars functors in the obvious sense (see in particular [MV07, Prop. 8.1] for the case of $\mathcal{J}_!(\lambda,\Bbbk)$; the case of the Whittaker standard object is much easier since no perverse truncation is involved). If $\Bbbk\in\{\mathbb{K},\mathbb{L}\}$, by Proposition 3.11 we know that $\Phi^0(\mathcal{J}_!(\lambda,\Bbbk))$ has no quotient of the form $\mathrm{IC}^{\mathcal{I}\mathcal{W}}_{\mu+\varsigma}(\Bbbk)$ with $\mu\neq\lambda$; therefore $f^\Bbbk_\lambda$ is surjective.
The surjectivity of $f^\mathbb{L}_\lambda$ implies that $f^\mathbb{O}_\lambda$ must be surjective also. On the other hand, by Remark 3.7(1) the object $\Delta^{\mathcal{I}\mathcal{W}}_{\lambda+\varsigma}(\mathbb{K})$ is simple; hence $f^\mathbb{K}_\lambda$ is injective, which implies that $\ker(f^\mathbb{O}_\lambda)$ is a torsion object. Since this object embeds in the torsion-free object $\Delta^{\mathcal{I}\mathcal{W}}_{\lambda+\varsigma}(\mathbb{O})$, it must be zero. We finally obtain that $f^\mathbb{O}_\lambda$ is an isomorphism, so that $f^\mathbb{K}_\lambda$ and $f^\mathbb{L}_\lambda$ are isomorphisms as well.
Once we know that $f^\Bbbk_\lambda$ is an isomorphism, by Verdier duality (see the comments preceding Lemma 3.8) we deduce that $g^\Bbbk_\lambda$ is an isomorphism as well. (More precisely, we use the claim about $f^\Bbbk_\lambda$ in the setting where ψ is replaced by ψ^{-1}, and the fact that $\mathbb{D}_{\mathrm{Gr}}(\mathcal{J}_!(\lambda,\Bbbk))=\mathcal{J}_*(\lambda,\Bbbk)$, see [MV07, Prop. 8.1(c)].)

Now we conclude the proof as follows. Since Φ^0 is exact, it induces a functor

$$D^{\mathrm{b}}(\Phi^0) : D^{\mathrm{b}}\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk) \to D^{\mathrm{b}}\mathrm{Perv}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr}, \Bbbk).$$

We will prove that $D^{\mathrm{b}}(\Phi^0)$ is an equivalence, which will imply that Φ^0 is an equivalence as well, hence will conclude the proof. It is not difficult to see that the category $D^{\mathrm{b}}\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$, resp. $D^{\mathrm{b}}\mathrm{Perv}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr}, \Bbbk)$, is generated as a triangulated category by the objects $\{\mathcal{J}_!(\lambda, \Bbbk) : \lambda \in \boldsymbol{X}^\vee_+\}$, resp. by the objects $\{\Delta^{\mathcal{I}\mathcal{W}}_{\lambda+\varsigma}(\Bbbk) : \lambda \in \boldsymbol{X}^\vee_+\}$, as well as by the objects $\{\mathcal{J}_*(\lambda, \Bbbk) : \lambda \in \boldsymbol{X}^\vee_+\}$, resp. by the objects $\{\nabla^{\mathcal{I}\mathcal{W}}_{\lambda+\varsigma}(\Bbbk) : \lambda \in \boldsymbol{X}^\vee_+\}$. Hence to conclude it suffices to prove that for any $\lambda, \mu \in \boldsymbol{X}^\vee_+$ and any $n \in \mathbb{Z}$ the functor Φ^0 induces an isomorphism

$$\mathrm{Ext}^n_{\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr},\Bbbk)}(\mathcal{J}_!(\lambda, \Bbbk), \mathcal{J}_*(\mu, \Bbbk)) \xrightarrow{\sim} \mathrm{Ext}^n_{\mathrm{Perv}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr},\Bbbk)}(\Delta^{\mathcal{I}\mathcal{W}}_{\lambda+\varsigma}(\Bbbk), \nabla^{\mathcal{I}\mathcal{W}}_{\mu+\varsigma}(\Bbbk)).$$

However, this is clear from Remark 3.2(2) and Remark 3.7(2). □

We continue with the assumptions of Sections 2–3; but from now on (except in Remark 4.17) for simplicity we assume that $\Bbbk$ is a field.

4.3. Tilting objects in the Satake category. Recall the notion of parity complexes in $D^{\mathrm{b}}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr}, \Bbbk)$ considered in Remark 3.5. In certain proofs of this subsection we will also consider the $(I^+_{\mathrm{u}}, \chi^*_{I^+}(\mathcal{L}^{\Bbbk}_{\psi}))$-equivariant constructible derived category of Fl, which we will denote $D^{\mathrm{b}}_{\mathcal{I}\mathcal{W}}(\mathrm{Fl}, \Bbbk)$. Of course, we can also consider parity complexes in this category, as well as in the I^--equivariant derived category $D^{\mathrm{b}}_{I^-}(\mathrm{Fl}, \Bbbk)$, or in the $G_{\mathscr{O}}$-constructible derived category $D^{\mathrm{b}}_{(G_{\mathscr{O}})}(\mathrm{Gr}, \Bbbk)$. (Note that, by definition, an object of $D^{\mathrm{b}}_{I^-}(\mathrm{Fl}, \Bbbk)$ is parity iff its image in the I^--constructible derived category $D^{\mathrm{b}}_{(I^-)}(\mathrm{Fl}, \Bbbk)$ is parity.) In particular, for any $\lambda \in \boldsymbol{X}^\vee_+$, we denote by $\mathcal{E}_\lambda$ the unique indecomposable parity complex in the category $D^{\mathrm{b}}_{(G_{\mathscr{O}})}(\mathrm{Gr}, \Bbbk)$ supported on $\overline{\mathrm{Gr}^\lambda}$ and whose restriction to Gr^λ is $\underline{\Bbbk}_{\mathrm{Gr}^\lambda}[\dim \mathrm{Gr}^\lambda]$ (see [JMW14, Th. 2.12 & §4.1]). Since $\mathsf{H}^\bullet_{G_{\mathscr{O}}}(\mathrm{pt}; \Bbbk)$ might not be concentrated in even degrees, in general the theory of [JMW14] does not apply in $D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$. This difficulty will be remedied by the following lemma.

Lemma 4.8. *Any parity object $\mathcal{E}$ in $D^{\mathrm{b}}_{(G_{\mathscr{O}})}(\mathrm{Gr}, \Bbbk)$ is a direct summand of a parity object $\mathcal{E}'$ which belongs to the essential image of the functor $\mathsf{For}_{G_{\mathscr{O}}} : D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk) \to D^{\mathrm{b}}_{(G_{\mathscr{O}})}(\mathrm{Gr}, \Bbbk)$.*

Proof. Of course we can assume that $\mathcal{E} = \mathcal{E}_\lambda$ for some $\lambda \in \boldsymbol{X}^\vee_+$. Recall that the forgetful functor For_{I^-} sends indecomposable parity objects to indecomposable parity objects (see [MR18, Lem. 2.4]). In view of the classification of such objects in the I^--equivariant and I^--constructible derived categories, this means that any I^--constructible parity complex on Gr belongs to the essential image of For_{I^-}. In particular, there exists a parity complex $\mathcal{F}$ in $D^{\mathrm{b}}_{I^-}(\mathrm{Gr}, \Bbbk)$ such that $\mathcal{E}_\lambda \cong \mathsf{For}_{I^-}(\mathcal{F})$. Now we set $\mathcal{E}' := \mathsf{For}_{G_{\mathscr{O}}}(^*\mathsf{Ind}^{G_{\mathscr{O}}}_{I^-}(\mathcal{F}))$. Then $\mathcal{E}'$ is parity as a convolution of parity complexes, see [JMW14, Th. 4.8]. And since this object is supported on $\overline{\mathrm{Gr}^\lambda}$ and has nonzero restriction to Gr^λ, it must admit a cohomological shift of $\mathcal{E}_\lambda$ as a direct summand. □

Recall that the forgetful functor $\mathsf{For}_{G_{\mathscr{O}}} : D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk) \to D^{\mathrm{b}}_{(G_{\mathscr{O}})}(\mathrm{Gr}, \Bbbk)$ restricts to an equivalence between $G_{\mathscr{O}}$-equivariant and $G_{\mathscr{O}}$-constructible perverse sheaves, see [MV07, Prop. 2.1] (or [BR18, Prop. 1.10.8]). Therefore for any $\mathcal{F}$ in $D^{\mathrm{b}}_{(G_{\mathscr{O}})}(\mathrm{Gr}, \Bbbk)$ and any $n \in \mathbb{Z}$, the perverse sheaf ${}^p\mathcal{H}^n(\mathcal{F})$ is $G_{\mathscr{O}}$-equivariant. The main result of this section is the following.

Theorem 4.10. *For any $n \in \mathbb{Z}$ and $\lambda \in \boldsymbol{X}^\vee_+$, the $G_{\mathscr{O}}$-equivariant perverse sheaf ${}^p\mathcal{H}^n(\mathcal{E}_\lambda)$ is tilting in the highest weight category $\mathrm{Perv}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$. In particular, the indecomposable tilting object associated with λ is a direct summand of ${}^p\mathcal{H}^0(\mathcal{E}_\lambda)$.*

The proof of Theorem 4.10 requires a few preliminaries. We start with the following observation, which will be crucial for us.

Proposition 4.12. *The parity objects in $D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$ are exactly the direct sums of cohomological shifts of tilting perverse sheaves.*

Proof. As already noticed in the proof of Corollary 3.6, the strata $X_\lambda \subset \mathrm{Gr}$ supporting $(I^+_{\mathrm{u}}, \chi^*_{I^+}(\mathcal{L}^{\Bbbk}_\psi))$-equivariant local systems (i.e., those with $\lambda \in \boldsymbol{X}^\vee_{++}$) have dimensions of constant parity on each connected component of Gr. Therefore, the tilting objects in the highest weight category $\mathrm{Perv}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$ are also parity. By unicity, they must then coincide with the "parity sheaves" (or, in another terminology, normalized indecomposable parity complexes) of [JMW14, Def. 2.14]. The claim follows, since any parity complex is a direct sum of cohomological shifts of such objects. □

Next we observe that the parity property is preserved under convolution, in the following sense.

Lemma 4.13. *If $\mathcal{F} \in D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Fl}, \Bbbk)$ and $\mathcal{G} \in D^{\mathrm{b}}_{I^-}(\mathrm{Fl}, \Bbbk)$ are parity complexes, then $\mathcal{F} \star^{I^-} \mathcal{G} \in D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Fl}, \Bbbk)$ is a parity complex.*

Proof. In view of the description of parity complexes in [JMW14, §4.1], the claim follows from standard arguments going back at least to [Spr82]. In fact it suffices to treat the case $\mathcal{G} = \underline{\Bbbk}_{\overline{\mathrm{Fl}_w}}$ when $\ell(w) \in \{0, 1\}$, which can be done "by hand" as in [Spr82]. □

Lemma 4.14. *If $\mathcal{F} \in D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$ is parity and $\mathcal{G} \in D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$ is such that $\mathsf{For}_{G_{\mathscr{O}}}(\mathcal{G})$ is parity, then $\mathcal{F} \star^{G_{\mathscr{O}}} \mathcal{G} \in D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$ is parity.*

Proof. The natural projection $\pi : \mathrm{Fl} \to \mathrm{Gr}$ (a smooth and projective morphism) is I^+_{u}-equivariant, hence induces functors

$$\pi^* : D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk) \to D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Fl}, \Bbbk), \qquad \pi_* : D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Fl}, \Bbbk) \to D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk).$$

The projection formula shows that $\mathcal{F}$ is a direct summand in $\pi_*(\pi^*\mathcal{F})$, and by Lemma 2.5 we have

$$\pi_*(\pi^*\mathcal{F}) \star^{G_{\mathscr{O}}} \mathcal{G} \cong \pi^*(\mathcal{F}) \star^{I^-} \mathsf{For}^{G_{\mathscr{O}}}_{I^-}(\mathcal{G}).$$

Hence to conclude it suffices to prove that $\pi^*(\mathcal{F}) \star^{I^-} \mathsf{For}^{G_{\mathscr{O}}}_{I^-}(\mathcal{G})$ is parity. However we have

$$\pi^*\big(\pi^*(\mathcal{F}) \star^{I^-} \mathsf{For}^{G_{\mathscr{O}}}_{I^-}(\mathcal{G})\big) \cong \pi^*(\mathcal{F}) \star^{I^-} \pi^*(\mathsf{For}^{G_{\mathscr{O}}}_{I^-}(\mathcal{G})).$$

Since $\pi^*\mathcal{F}$ and $\pi^*(\mathsf{For}^{G_{\mathscr{O}}}_{I^-}(\mathcal{G}))$ are parity (because π is smooth), Lemma 4.13 implies that $\pi^*\big(\pi^*(\mathcal{F}) \star^{I^-} \mathsf{For}^{G_{\mathscr{O}}}_{I^-}(\mathcal{G})\big)$ is parity. We deduce that $\pi^*(\mathcal{F}) \star^{I^-} \mathsf{For}^{G_{\mathscr{O}}}_{I^-}(\mathcal{G})$ is parity, as expected. □

Corollary 4.15. *Let $\mathcal{E}$ be in $D^{\mathrm{b}}_{G_{\mathscr{O}}}(\mathrm{Gr}, \Bbbk)$, and assume that $\mathsf{For}_{G_{\mathscr{O}}}(\mathcal{E})$ is parity. Then $\Phi(\mathcal{E})$ is parity in $D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$. In particular, $\Phi^0({}^p\mathcal{H}^n(\mathcal{E}))$ is a tilting perverse sheaf for any $n \in \mathbb{Z}$.*

Proof. Since $\Delta^{\mathcal{IW}}_{\varsigma}(\Bbbk)$ is parity (see (3.2)), the first claim follows from Lemma 4.14. The second claim follows from the facts that Φ is t-exact and that the perverse cohomology objects of parity objects in $D^{\mathrm{b}}_{\mathcal{IW}}(\mathrm{Gr}, \Bbbk)$ are tilting perverse sheaves, see Proposition 4.12. □

We can finally give the proof of Theorem 4.10.

Proof of Theorem 4.10. Since Φ^0 is an equivalence of highest weight categories, to prove the first claim it suffices to prove that $\Phi^0({}^p\mathcal{H}^n(\mathcal{E}_\lambda))$ is tilting in the highest weight category $\mathrm{Perv}_{\mathcal{I}\mathcal{W}}(\mathrm{Gr}, \Bbbk)$. This follows from Lemma 4.8 and Corollary 4.15, since a direct summand of a tilting perverse sheaf is tilting. The second claim follows, since ${}^p\mathcal{H}^0(\mathcal{E}_\lambda)$ is supported on $\overline{\mathrm{Gr}^\lambda}$, and has nonzero restriction to Gr^λ. □

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